

DATE : 19.11.2025

DT/NT : DT

LESSON : STATISTICS I

**SUBJECT: MEASURE OF CENTRAL
TENDENCY**

BATCH : 322

**DATA
SCIENCE**



TECHPRO
EDUCATION



techproeducation.com



+1 (585) 304 29 59





Can you write a sentence that you remember from the previous lesson?



**Have you practiced
with the **pre-class**
materials that will
prepare you for
today's lesson?**

Session - 3 Content



**What will we
learn today?**

Measure of Central Tendency

- Mean
- Median
- Mode

Measure of Dispersion

- Range
- Variance
- Standard Deviation
- IQR

Content

➤ Central Tendency (Measure of Centre)

- Mean
- Median
- Mode



➤ Dispersion (Measure of Spread)

- Range
- IQR
- Variance
- Standard Deviation

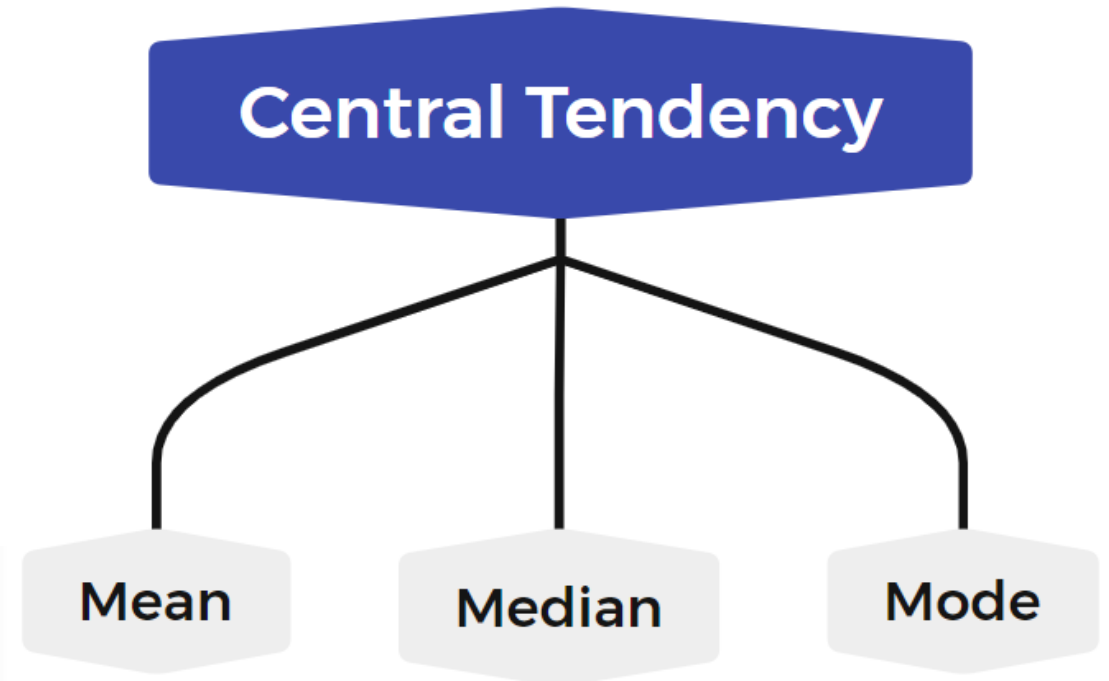
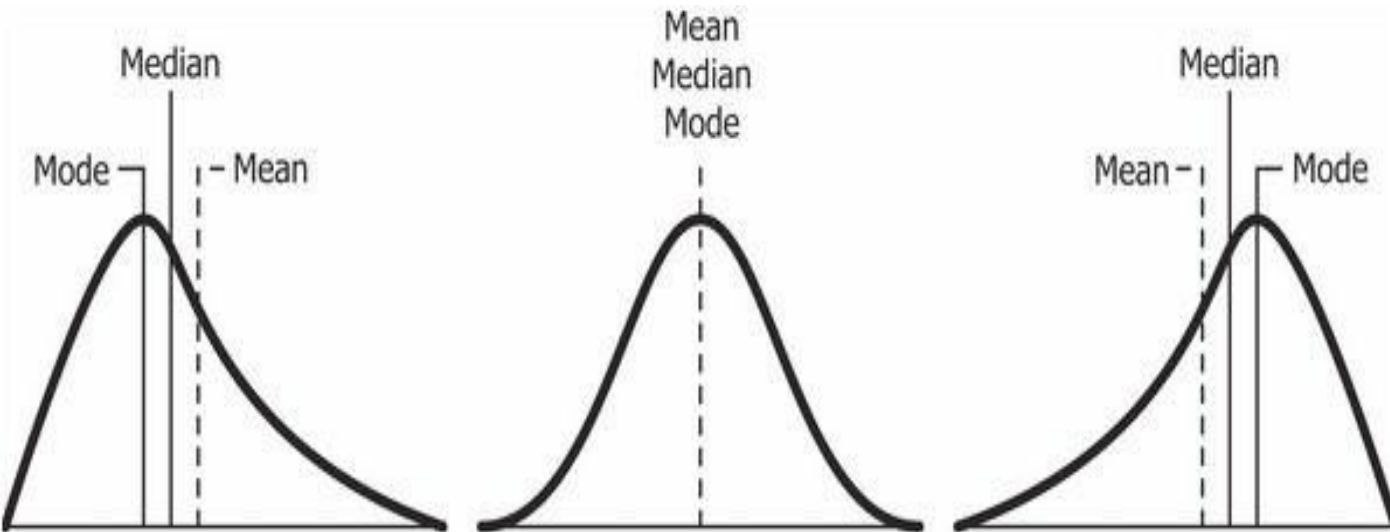


Central Tendency (Measure of Centre)

- **Mod**
- **Median**
- **Mean**

Central Tendency

- ▶ Best description of data with a single value;
 - **Mean:** Average, arithmetic mean
 - **Median:** Central point
 - **Mode:** Highest frequency



Mean (Average)

- Divide the sum of the data by the total number of observations
- Used to determine the location of the distribution

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^N x_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$
N = number of items in the population	n = number of items in the sample



We use “**mean**” for filling null values in “**numeric columns**”.



Mean Example

Example - 1

73	73	76	77	81	100
----	----	----	----	----	-----

$$\text{Mean (average)} = \frac{\text{Sum}}{\text{Count}}$$

$$= \frac{73 + 73 + 76 + 77 + 81 + 100}{6}$$

$$= \frac{480}{6}$$

$$= 80$$

Example - 2

If we have frequencies, how can we calculate mean?

x	frequency
10	3
12	5
15	2
17	6
20	1
24	4

$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i}$$

$$\bar{x} = \frac{10 \times 3 + 12 \times 5 + 15 \times 2 + 17 \times 6 + 20 \times 1 + 24 \times 4}{3 + 5 + 2 + 6 + 1 + 4}$$

$$\bar{x} = \frac{338}{21}$$

$$\bar{x} = 16.095$$

Mean (Average)

Staff	Salary (thousand \$)
1	102
2	33
3	26
4	27
5	30
6	25
7	33
8	33
9	24

Mean = 37

But except the outlier (102), none of them not close to 37. Outlier caused a misinterpretation. Mean cannot represent the data set while there is outlier in the data set.

Staff	Salary (thousand \$)
2	33
3	26
4	27
5	30
6	25
7	33
8	33
9	24

Mean = 28.9

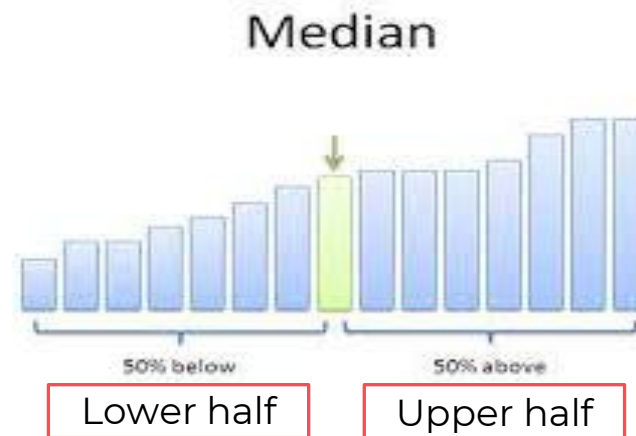
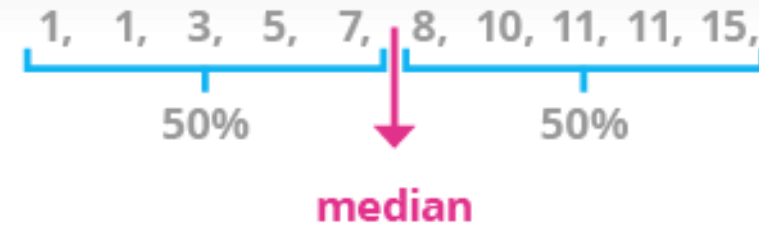
When we eliminate the outlier, mean is 28.9. It is located the centre of data set. It can represent the data set now.

Median

- It is the median (**central**) **score of a data set sorted from smallest to largest**.
- If the number of data is odd, the median is **1** value, but if the number is even, the average of the middle **2** values is taken to find the median.

Median Formula $\left(\frac{n+1}{2}\right)^{\text{th}}$

- The median is the middle score. If the sample size is 9, the fifth element is the median



Staff	Salary (thousand \$)
1	24
2	25
3	26
4	27
5	30
6	33
7	33
8	33
9	102

Median Example

• Ex-1

Find the median of these ages.

53 32 61 57 39 44 57

Let's sort the numbers as ascending.

32 39 44 **53** 57 57 61

Median is 53.

• Ex-2



\$4000



\$15.000



HONDA

\$20.000



Mercedes-Benz

\$33.000



\$1.800.000

Mean:

$$\mu = \frac{\sum X}{N}$$

$$\mu = \frac{\$4000 + \$15000 + \$20000 + \$33000 + \$1800000}{5}$$

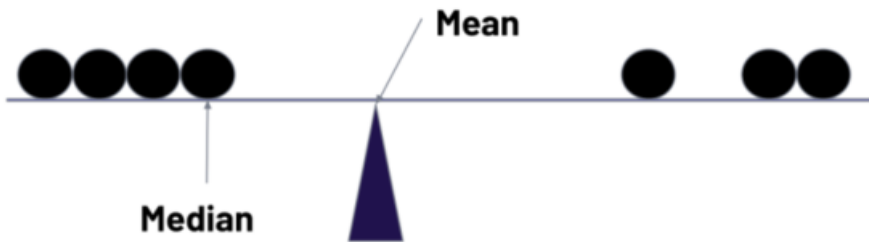
$$\mu = \frac{\$1872000}{5} = \$374400$$

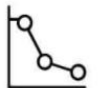
Median:

\$20000

Mean vs. Median

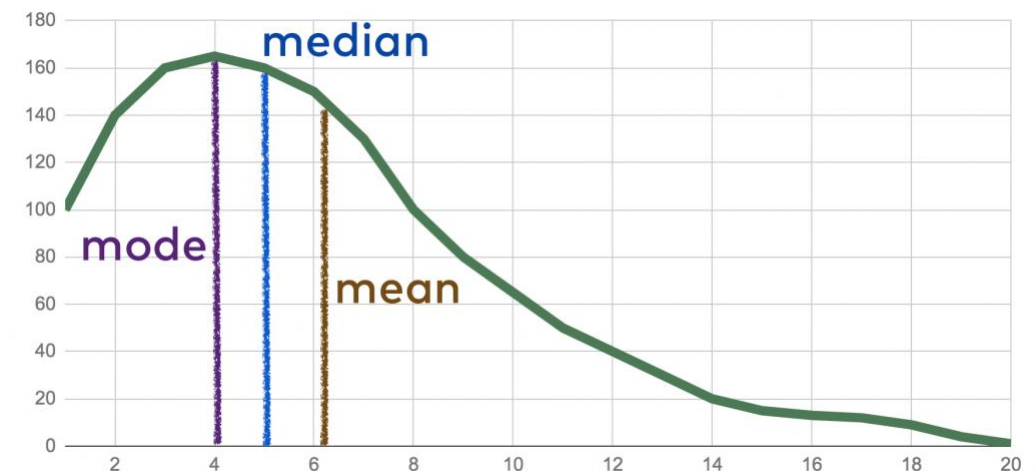
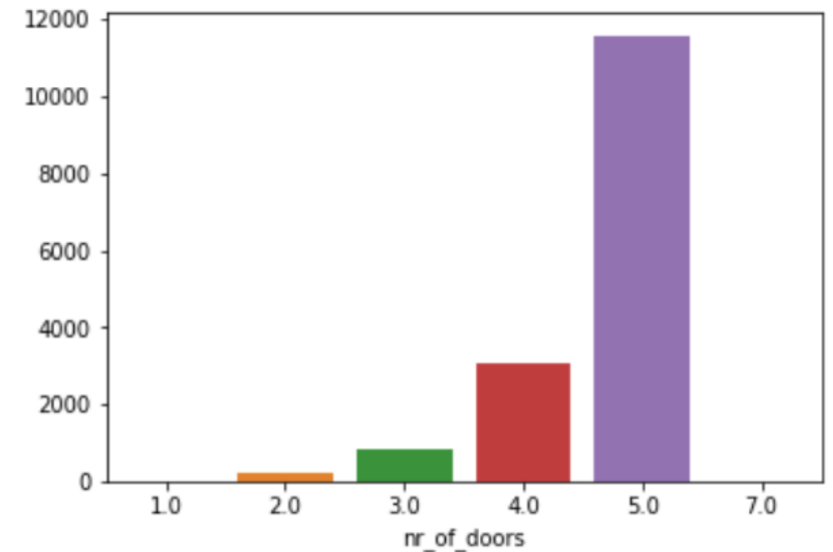
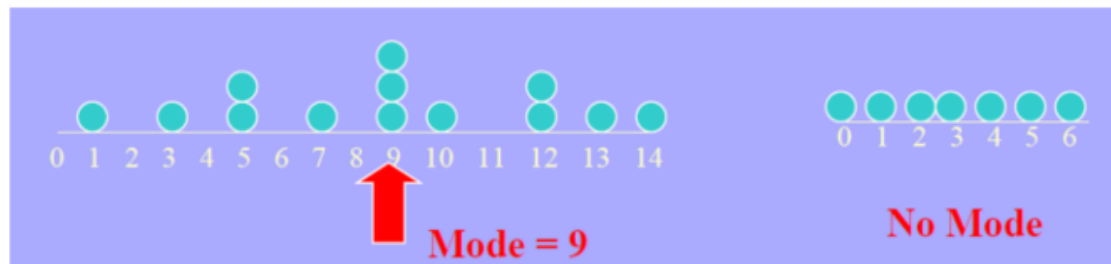
- If there are outliers in a small set of scores, the median is better.
- If there are no outliers in large data sets, the mean is better.
- Median may be better for salary bidding.



#1. Basic Definition	
Mean  Mean can be referred to the simple average or arithmetic average of the given set of data or the quantities or the values.	Median  Median can be defined as the middle most numerical in an ordered list (i.e. from lowest to highest or vice versa) of values.
#4. Calculation	
Mean  Mean can be calculated by adding up or taking up the sum of all the observations or the data set and then dividing that summation or the value obtained by the number of observations in the sample provided.	Median  To calculate median, first one needs to be arranged the data set in an ascending or in a descending order and then the value which shall fall in the exact middle of the new data set or of the sample, will be the median.
#2. Actually meaning	
Mean  Mean can also be termed as arithmetic average.	Median  Median can be meant as an positional average.
#5. What does it represent	
Mean  Mean will represent the central gravity of the data set given.	Median  The mid-point of the data set will be represented by the median.
#3. Type of distribution	
Mean  For Mean, normal distribution would apply.	Median  For median to be used and to be find it as more appropriate to use than mean, there should be skewed distribution.
#6. Outliners bias	
Mean  Mean will represent the central gravity of the data set given.	Median  The mid-point of the data set will be represented by the median.

Mode

- **Mode** is called **peak value**
- Mode: The most popular, most frequently encountered value in the data set.
- It can be used for both numeric and categorical variables
- Advantages and Disadvantages



Mode Example

• Ex-1

What is the mode of these numbers?

53 32 61 **57** 39 44 **57**

Mode is 57 since there are two 57.

• Ex-2

What is the mode of these numbers?

120 100 130 100 160 130 86 100 94 90

Let's sort the numbers as ascending;

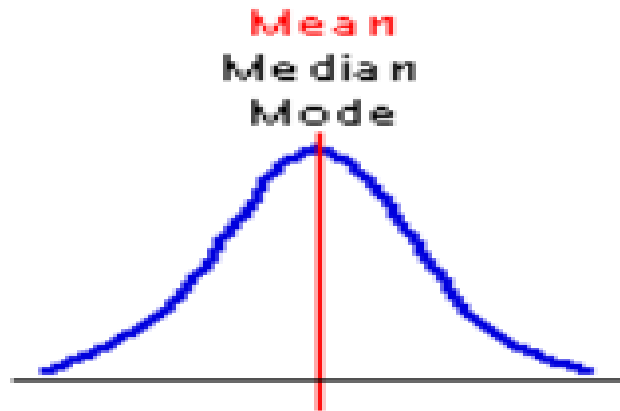
86	90	94	100	100	100	120	130	130	160
----	----	----	-----	-----	-----	-----	-----	-----	-----

Mode = 100

Mode is 100, because it has most frequency.

Mean, Median and Mode

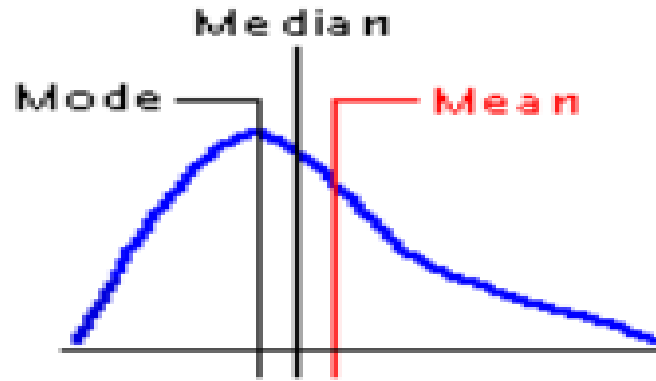
Normal Symmetric Distribution



Symmetrical
Distribution

Mean=Median=Mode

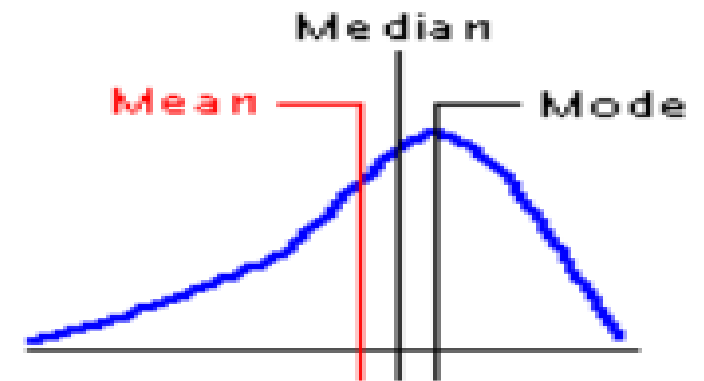
Right-Skewed Distribution



Positive
Skew

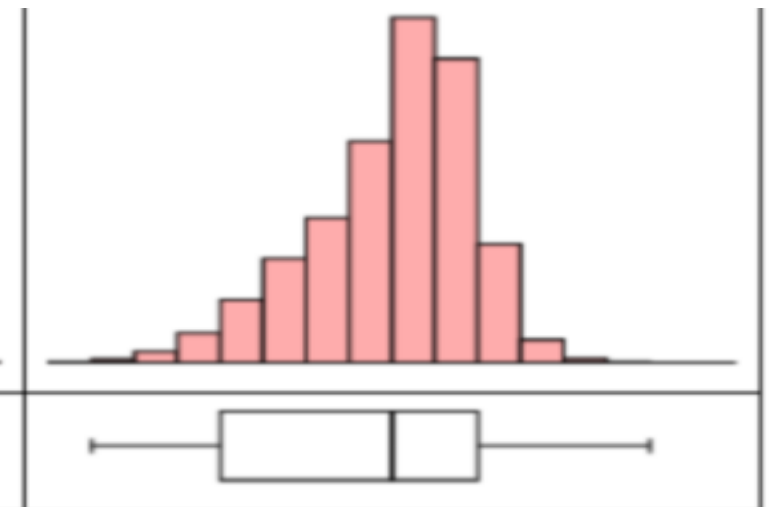
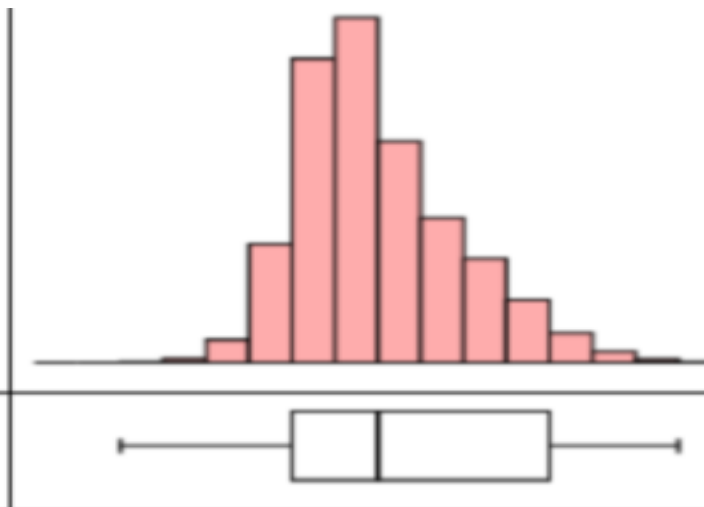
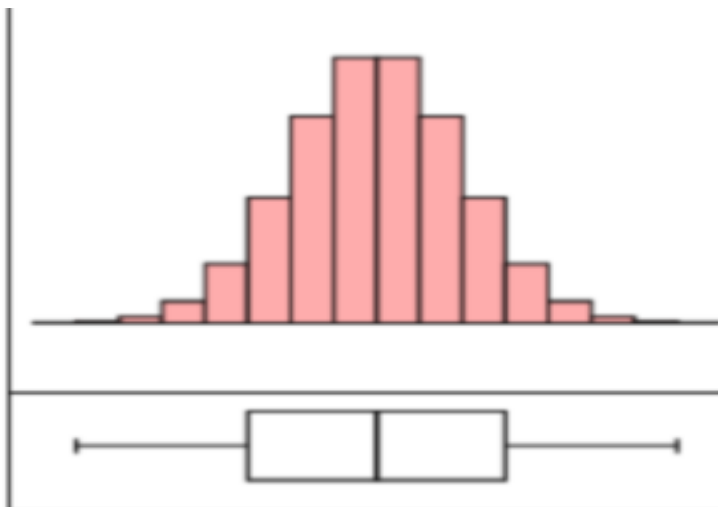
Mode < Median < Mean

Left-Skewed Distribution



Negative
Skew

Mean < Median < Mode



Top 60 Statistics Interview Questions 2024



Question 4: What is the relationship between mean and median in normal distribution?

Answer: In a normal distribution, the mean and the median are equal.

Top 60 Statistics Interview Questions 2024



Question 5: What is the left-skewed distribution and the right-skewed distribution?

Answer: In the left-skewed distribution, the left tail is longer than the right side. It is also known as **negative-skew** distribution.

Mean < median < mode

In the right-skewed distribution, the right tail is longer. It is also known as **positive-skew** distribution.

Mode < median < mean

Mode – Median-Mean Example



MEASURES OF CENTRE

MODE

MEDIAN

MEAN



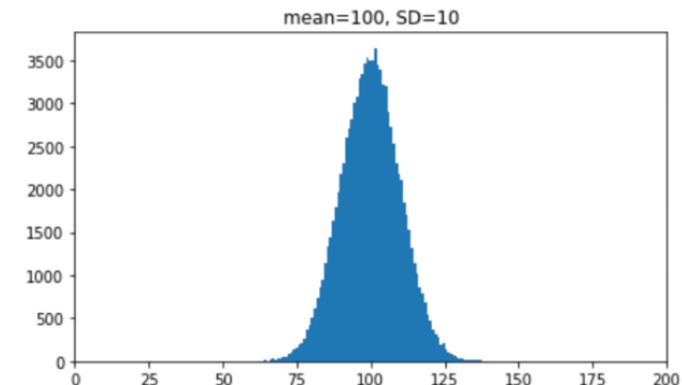
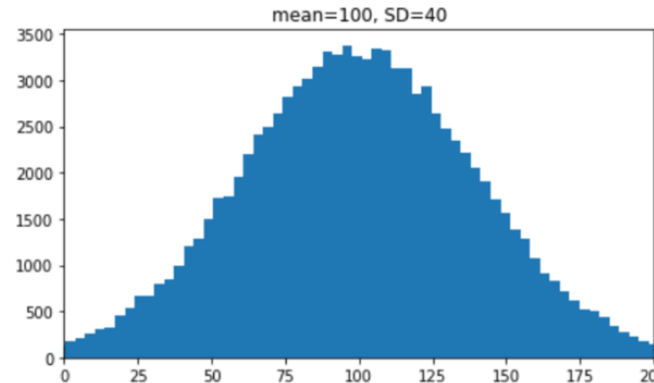
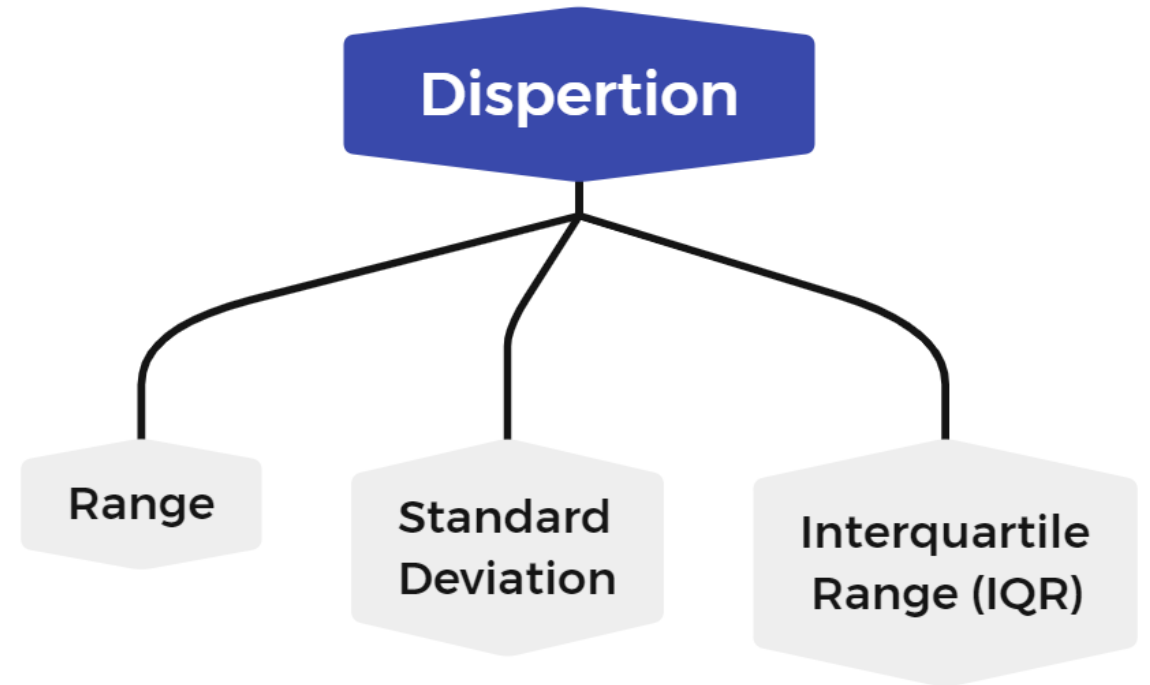
We use “**mode**” for filling null values in “**categorical columns**”.



Dispersion (Measure of Spread)

► Dispersion

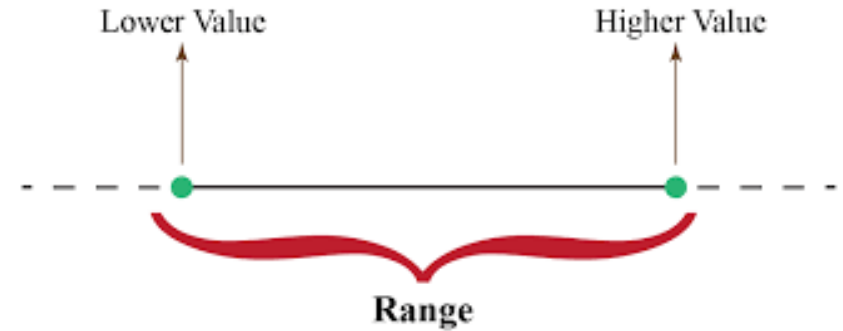
- Central tendency measures alone do not characterize the distribution.
- The fact that **the mean of two data groups is equal** does not **require that their distributions are the same**.
- **A distribution explains more than the central tendency** does.



Range

Range - Span - Width of Variation

- The **range** of a data set is the **difference** between the **maximum** and **minimum** number of entries in the set.
- It is the simplest measure of variability.



2 4 9 5 7 3

$$\text{Range} = \text{Largest} - \text{Smallest} = 9 - 2 = 7$$

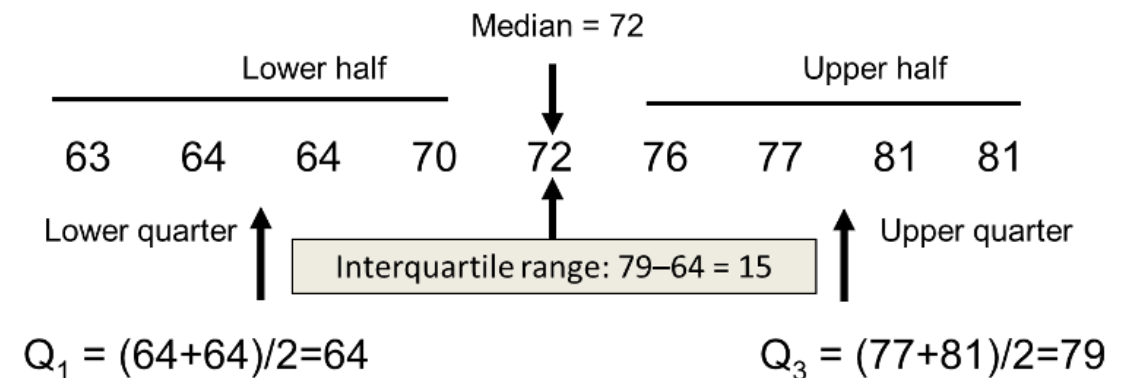
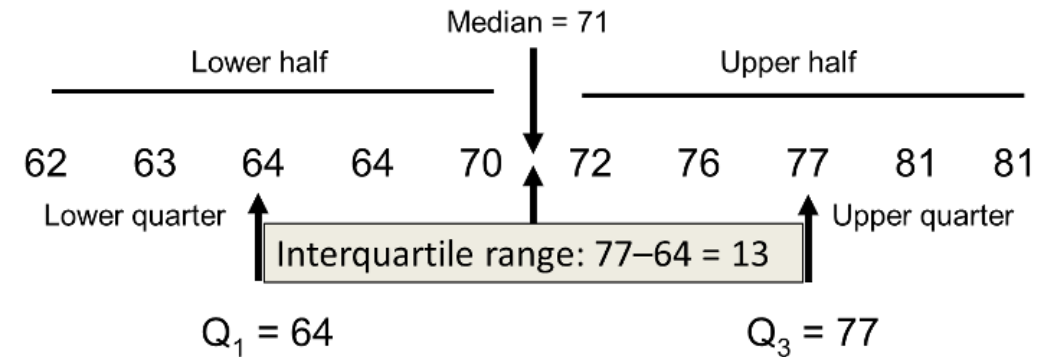
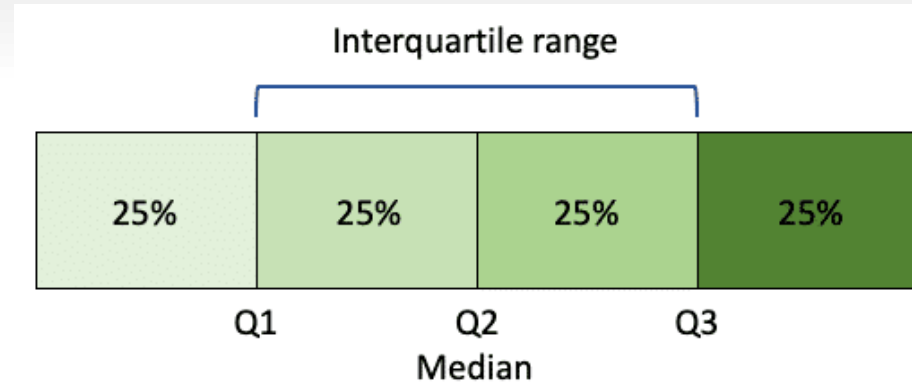
Recap Mean-Median-Mode-Range

1. Find the mean, median, mode, and range of the following data set: 10, 7, 14, 23, 15, 7, 32.

Inter Quartile Range (IQR)

- ▶ Quartiles are the values that divide a group of numbers by four parts.
- Q2 is the median of the whole dataset
- Q1 is the median of the portion below the median
- Q3 is the median of the portion above the median.

$$\text{IQR} = Q3 - Q1$$



IQR Example



There are six figures.

There are six figures.

Median



Q1

$$Q1 = (5+7)/2 = 6$$



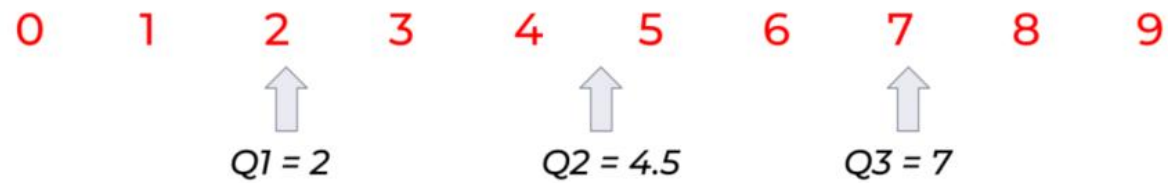
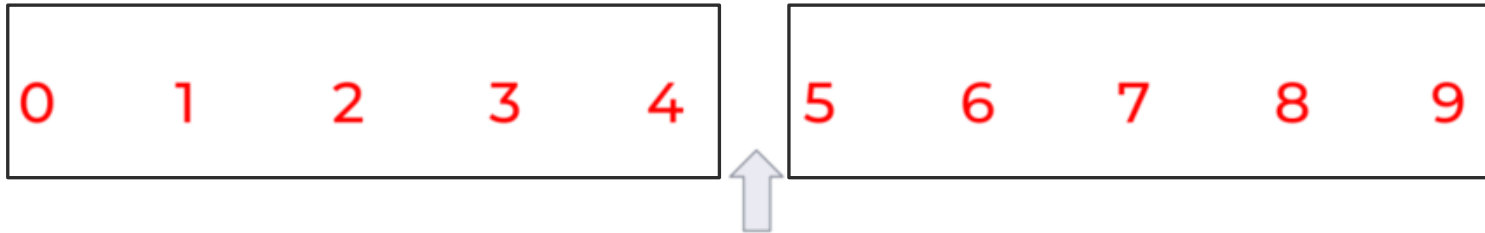
Q3

$$Q3 = (14+16)/2 = 15$$

$$IQR = Q3 - Q1$$

$$IQR = 15 - 6 = 9$$

IQR Example - 2



$$\text{Interquartile Range} = 7 - 2$$
$$\text{IQR} = 5$$



$$\text{Interquartile Range} = 7 - 2$$
$$\text{IQR} = 5$$

IQR Example – 2 - on Video



Top 60 Statistics Interview Questions 2024



Question 6: How to calculate range and interquartile range?

Answer: The range is the difference between the highest and the lowest values whereas the Interquartile range is the difference between upper and lower medians.

$$\text{Range (X)} = \text{Max(X)} - \text{Min(X)}$$

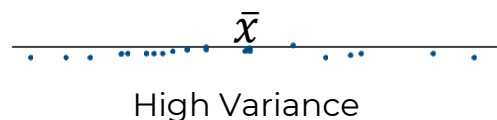
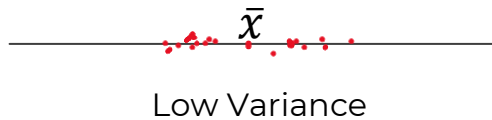
$$\text{IQR} = Q3 - Q1$$

Here, Q3 is the third quartile (75 percentile)

Here, Q1 is the first quartile (25 percentile)

Variance

- **Variance** is defined as the **mean of the squares** of the **differences from the mean**.
- The amount by each score moves away from the mean.



Variance

<u>Sample variance</u>	<u>Population variance</u>
$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$ S^2 = sample variance x_i = value of <u>i</u> th element $\bar{x}$ = sample mean n = sample size	$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$ σ^2 = population variance x_i = value of <u>i</u> th element μ = population mean N = population size

sample variance

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

number of observations

variance

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

number of elements

Variance Example

- Variance for the following 4 numbers

0 1 5 6

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

0 1 5 6

Mean: $\mu = \frac{\sum X}{N} = \frac{0+1+5+6}{4} = \frac{12}{4} = 3$

Dev Sum of Squares: $SS = \sum (X - \mu)^2$

$$SS = (0 - 3)^2 + (1 - 3)^2 + (5 - 3)^2 + (6 - 3)^2$$
$$SS = 9 + 4 + 4 + 9 = 26$$

Variance: $\sigma^2 = \frac{\sum (X - \mu)^2}{N}$

$$\sigma^2 = \frac{26}{4} = 6.5$$

- Ex- 2

10 12 17 20 25 27 42 45

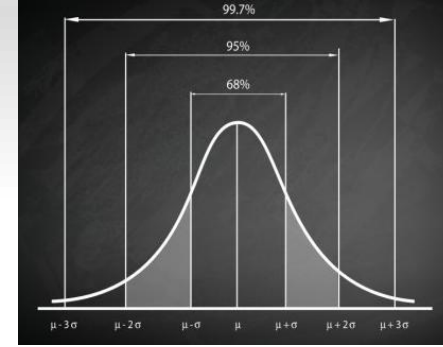
- Find for both sample and population.



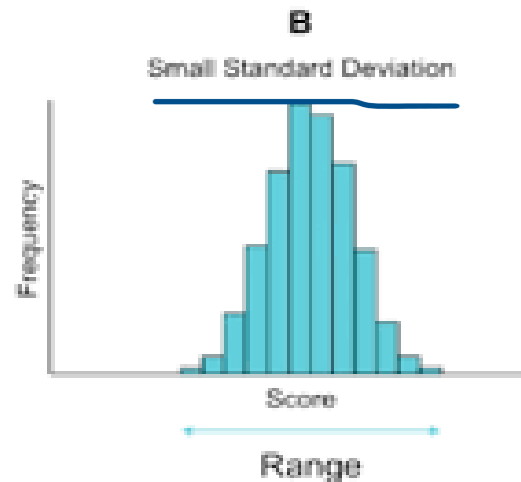
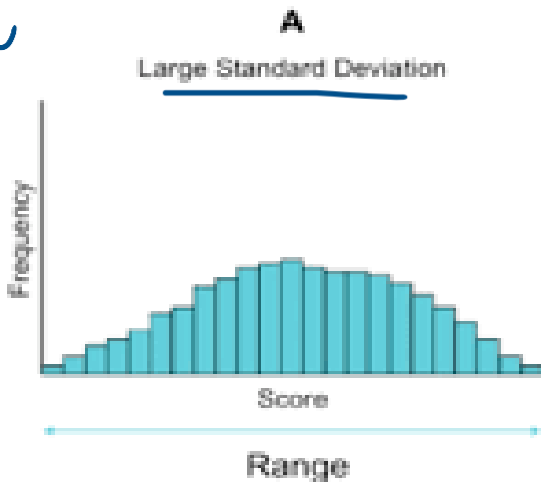
$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Standard Deviation



- It is the square root of the variance.
- The more spread out the data, the larger the standard deviation.
- If std dev is large, data is spread over an extensive range.



Sample

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Population

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

standard deviation $\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$

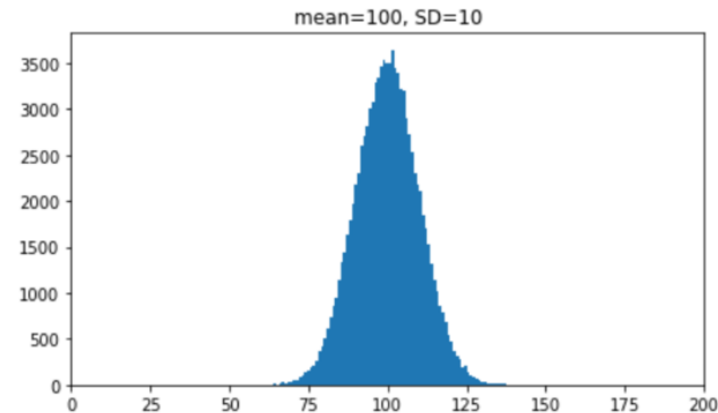
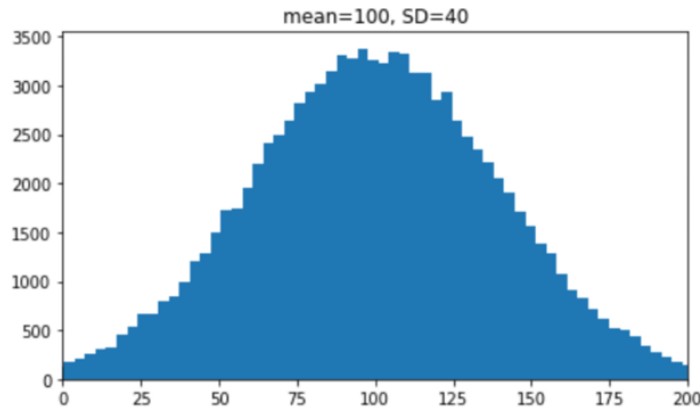
element x mean μ

number of elements N

Standard Deviation

While means equal in both data set, the first one has bigger std dev since it spread in a large range.

The second one has smaller std dev since it spread in a narrow range.



Std. Dev. Example

Staff	Salary (thousand \$)
1	24
2	25
3	26
4	27
5	30
6	33
7	33
8	33
9	102

$$\mu = \frac{24+25+26+27+30+33+33+33+102}{9}$$

$$\mu = \frac{333}{9} = 37$$

$$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$$

$$\sigma = \sqrt{\frac{(24-37)^2 + (25-37)^2 + (26-37)^2 + (27-37)^2 + (30-37)^2 + (33-37)^2 + (33-37)^2 + (33-37)^2 + (102-37)^2}{9}}$$

$$\sigma = \sqrt{\frac{(-13)^2 + (-12)^2 + (-11)^2 + (-10)^2 + (-7)^2 + (-4)^2 + (-4)^2 + (-4)^2 + (65)^2}{9}}$$

$$\sigma = \sqrt{\frac{169+144+121+100+49+16+16+16+4225}{9}}$$

$$\sigma = \sqrt{\frac{4856}{9}}$$

$$\sigma = \sqrt{539,55}$$

$$\sigma = 23,22833518$$

Staff	Salary (thousand \$)
1	24
2	25
3	26
4	27
5	30
6	33
7	33
8	33

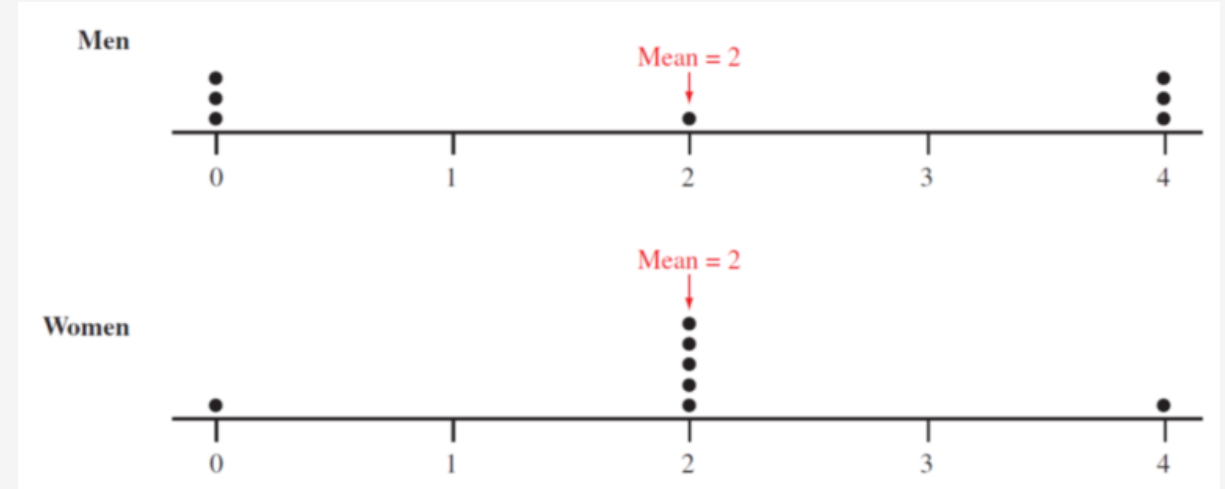
$$\sigma = 3.58$$

Std. Dev. Example - 2

Men: 0 0 0 2 4 4 4

Women: 0 2 2 2 2 2 4

- For the above 2 groups of respondents who answered the ideal number of children for a family (7 people each).
- What is the variance



Men:
$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{24}{6}} = \sqrt{4} = 2.0.$$

Women:
$$s = 1.2$$

RANGE STANDARD DEVIATION



Top 60 Statistics Interview Questions 2024



Question 7: What are descriptive statistics?

Answer: Descriptive statistics are used to summarize the basic characteristics of a data set in a study or experiment. It has three main types

- **Distribution:** Refers to the frequencies of responses. *Example:* Mode
- **Central Tendency:** Gives a measure or the average of each response. *Example:* Mean, median.
- **Variability:** Shows the dispersion of a data set. *Example:* Variance, standard deviation, range, IQR

Top 60 Statistics Interview Questions 2024



Question 8: What is the meaning of standard deviation?

Answer: Standard deviation gives the measure of the variation of dispersion of values in a data set. It represents the differences of each observation or data point from the mean.

$$(\sigma) = \sqrt{(\sum (x - \mu)^2 / n)}$$

Where the variance is the square of standard deviation.

Top 60 Statistics Interview Questions 2024



Question 9: What is Bessel's correction?

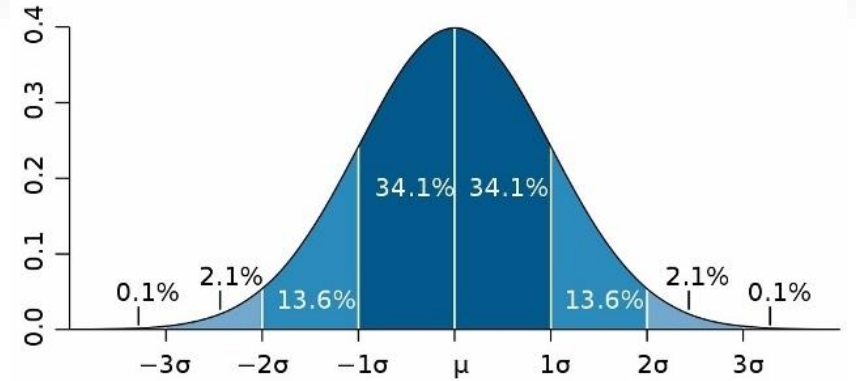
Answer: Bessel's correction advocates the use of **n-1** instead of **n** in the formula of standard deviation at **sample**. It helps to increase the accuracy of results while analyzing a sample of data to derive more general conclusions.

<u>Sample</u>	<u>Population</u>
$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$	$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$

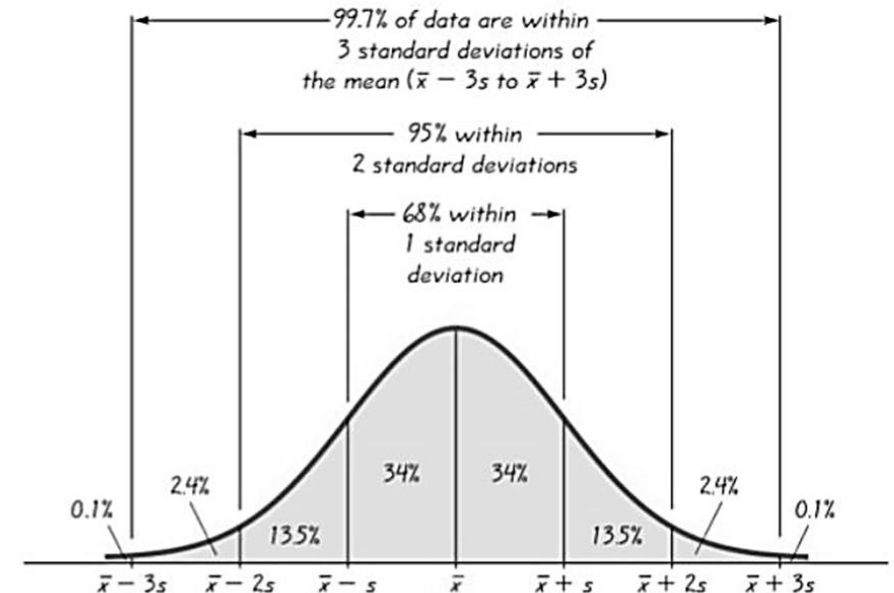
Empirical Rule

3 Sigma Rule

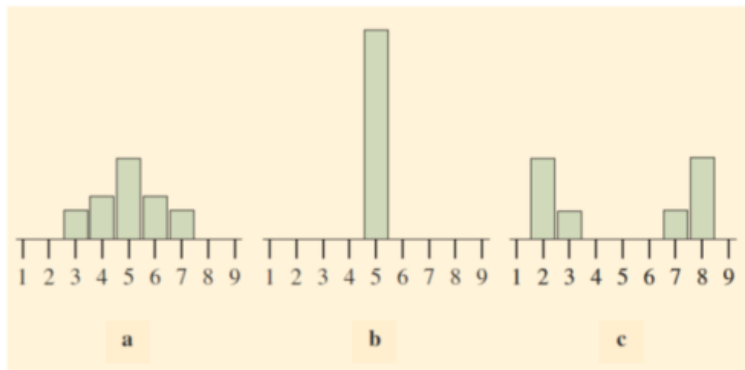
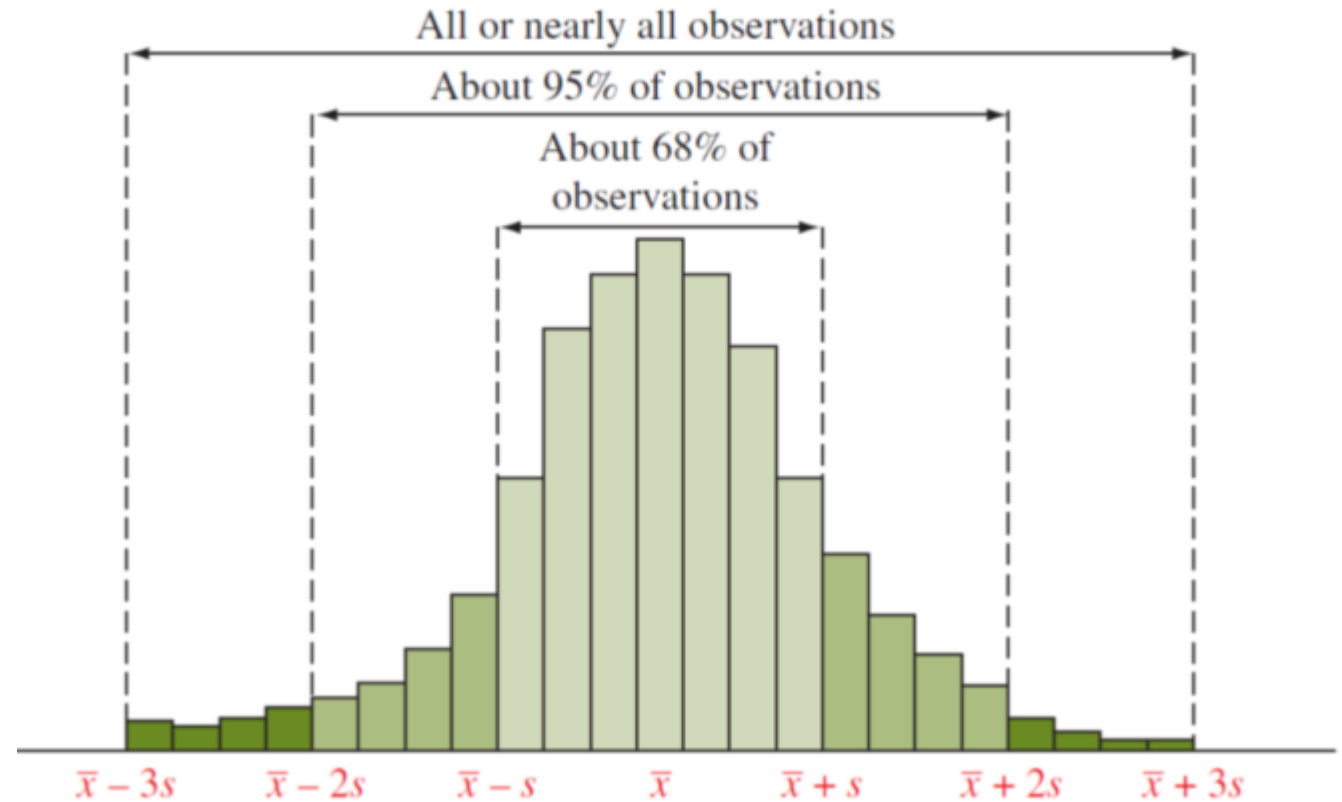
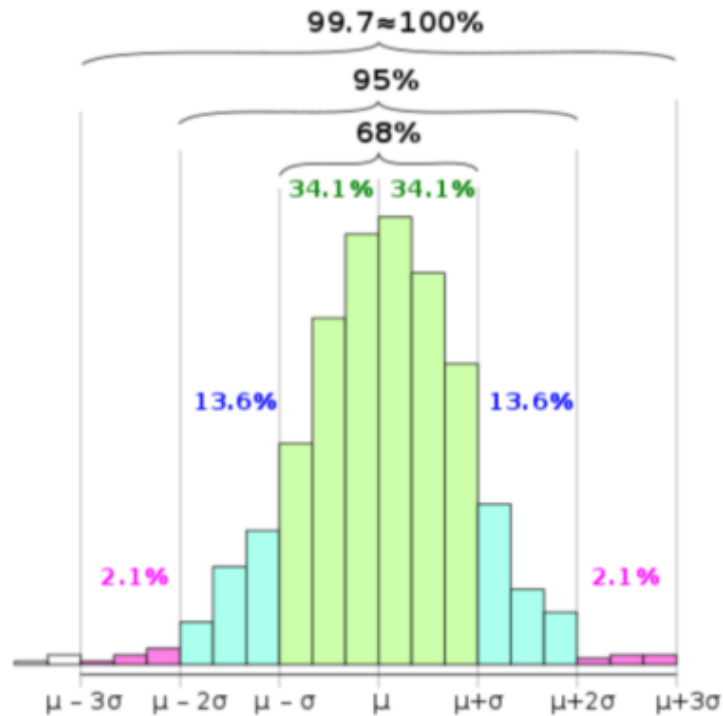
- Three Sigma Rule or also known as 68-95-99.7 Rule.
- Empirical Rule**
 - rule at 68% = (Mean - standard deviation) and (Mean + standard deviation)
 - rule at 95% = (Mean - 2 × standard deviation) and (Mean + 2 × standard deviation)
 - rule at 99.7% = (Mean - 3 × standard deviation) and (Mean + 3 × standard deviation)



The Empirical Rule

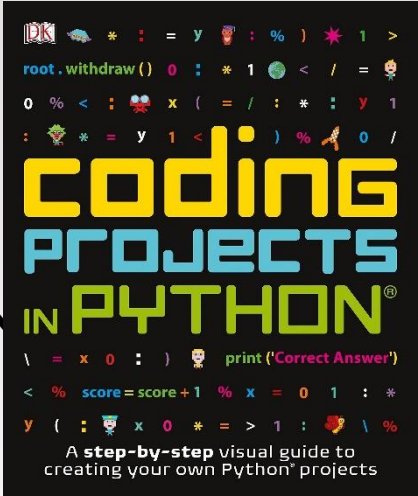


Empirical Rule

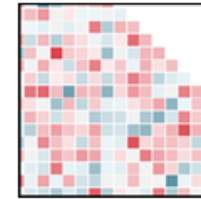


Statistics Practice-1

Python Notebook Time

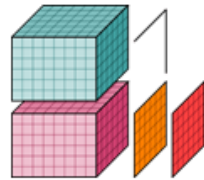


Statistics with Python



Seaborn

pandas
 $y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$



xarray



scikit-image
image processing in python



NumPy



SciPy



matplotlib



jupyter



IP[y]:
IPython



statsmodels

Statistic with Python

• Input

```
import numpy as np
from scipy import stats

salary = [102, 33, 26, 27, 30, 25, 33, 33, 24]

mean_salary = np.mean(salary)
print("mean:", mean_salary)

median_salary = np.median(salary)
print("median:", median_salary)

mode_salary = stats.mode(salary)
print("mode:", mode_salary)
```

• Output

```
mean: 37.0
median: 30.0
mode: ModeResult(mode=array([33]), count=array([3]))
```

Calculate Mean, Median and Mode with Python

Std. Dev with python

input :

```
import numpy as np

salary = [102, 33, 26, 27, 30, 25, 33, 33, 24]

print("Range: ", (np.max(salary)-np.min(salary)))

print("Variance: ", (np.var(salary)))

print("Std: ", (np.std(salary)))
```

output :

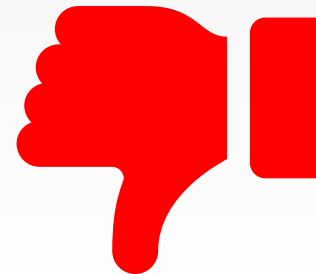
```
Range:  78
Variance:  539.5555555555555
Std:  23.22833518691246
```

It is time to Kahoot !!

Kahoot!



**Today's session was
effective/fruitful!!**



10:00

Start	Stop	Reset	mins: 10	secs: 0	type: None ▼
-------	------	-------	---------------------------------------	--------------------------------------	---